The proof is an immediate consequence of the linearity of the divergence operator, since the Liouville operator is constructed from this.

8.2.2 Nosé-Hoover Chains

Let us employ the additivity property to derive "Nosé-Hoover Chains" [251], a popular deterministic thermostat designed to give a more thorough sampling of the phase space than the standard Nosé-Hoover method.

The equations of Nosé-Hoover Chains may be written [251]

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \tag{8.16}$$

$$\dot{\boldsymbol{p}} = -\nabla U(\boldsymbol{q}) - \xi_1 \boldsymbol{p}, \tag{8.17}$$

$$\mu_1 \dot{\xi}_1 = \boldsymbol{p}^T \boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T} - \mu_1 \xi_2 \xi_1, \tag{8.18}$$

$$\mu_2 \dot{\xi}_2 = \mu_1 \xi_1^2 - \mathrm{k_B T} - \mu_2 \xi_2 \xi_3, \tag{8.19}$$

$$\mu_3 \dot{\xi}_3 = \mu_2 \xi_2^2 - \mathrm{k_B T} - \mu_3 \xi_3 \xi_4, \tag{8.20}$$

$$\vdots$$

$$\mu_k \dot{\xi}_k = \mu_{k-1} \xi_{k-1}^2 - \mathrm{k_B T}. \tag{8.21}$$

By creating a more substantial bath of oscillators, each of which is forced to have the target kinetic energy, the physical system may more closely approximate the canonical distribution.

To see that this is a thermostat, note the equations (8.16)–(8.21) can be split into the following pieces:

$$\begin{bmatrix} \dot{\boldsymbol{q}} \\ \dot{\boldsymbol{p}} \\ \mu_1\dot{\xi}_1 \\ \mu_2\dot{\xi}_2 \\ \vdots \\ \mu_{k-1}\dot{\xi}_{k-1} \\ \mu_k\dot{\xi}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{p} \\ -\nabla U(\boldsymbol{q}) - \xi_1\boldsymbol{p} \\ \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T} \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ -\mu_1\xi_1\xi_2 \\ \mu_1\xi_1^2 - \mathrm{k_B T} \\ \vdots \\ 0 \\ 0 \end{bmatrix} + \cdots + \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{0} \\ 0 \\ 0 \\ \vdots \\ -\mu_{k-1}\xi_{k-1}\xi_k \\ \mu_{k-1}\xi_{k-1}^2 - \mathrm{k_B T} \end{bmatrix}.$$

The first piece of the system corresponds to Nosé-Hoover, which we already know preserves $\rho_\beta \exp(-\beta\mu_1\xi_1^2/2)$. By examination, the distribution $\exp(-\beta\mu_1\xi_1^2/2)$ $\exp(-\beta\mu_2\xi_2^2/2)$ is preserved by the second piece, the next piece preserves

$\exp(-\beta\mu_2\xi_2^2/2)\exp(-\beta\mu_3\xi_3^2/2)$ and so on. Thus the full system preserves the combined distribution

$$\tilde{\rho} = \rho_\beta \exp\left(-\beta \sum_{i=1}^{k} \mu_i \xi_i^2/2\right).$$

Of course this is only a demonstration that the dynamics is compatible with $\tilde{\rho}$; there may be other invariant distributions.

Numerical treatment of the Nosé-Hoover Chain system can be performed by splitting, just as in Newtonian and Langevin dynamics. Various related methods and comparisons may be found in [65, 192, 230, 253].

8.2.3 Lack of Ergodicity of Deterministic Thermostats

Deterministic methods like the ones in the previous subsection are sometimes used for sampling calculations, however, our and others' experience is that these methods are not always ergodic.

In the case of Nosé-Hoover, this problem was known to both of the inventors and has been observed by many authors (see e.g. [251] and [180]). More recently, it has been studied analytically [218]. Some confusion arises from the fact that the practical ergodic properties (and other properties) in Nosé-Hoover simulations are sensitive to the particular system under study and the values of parameters. For example it is known that in the case of simple liquids and gases, Nosé-Hoover does (at least under some conditions) enable reliable calculation of thermodynamical quantities [77], whereas in other systems results are poor and cannot be rectified by adjustment of parameters. In general, the control of temperature requires that the thermostat couples tightly to the physical variables, which generally calls for a small thermal mass, but this may be at odds with efficiency reduction (it may lead to a need to use smaller timesteps to resolve the motion of the auxiliary variable).

Ergodicity is less well known as a potential problem for methods involving thermostatting chains. Figure 8.1 shows a projection of a torus in phase space found numerically for a chain-thermostatted harmonic oscillator. As the number of auxiliary chain variables increases, we observe that the tori become increasingly thin but wind in a complicated way through the phase space. Even for initial conditions trapped within one of the tori, crude sampling may be possible, provided that the torus itself is of sufficient size, but the presence of such tori will impose a finite accuracy limit on the averages that can be obtained.

The reliable use of deterministic thermostats is system dependent—it is essential that there be mechanisms within the underlying physical dynamics model that promote energetic exchange among the degrees of freedom.

Numerical methods for Nosé-Hoover chains are easily constructed by splitting of the equations of motion.